

# Violet Xiang

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## Education

### Stanford University

*Ph.D. in Psychology*

Research focus: LLM post-training, reinforcement learning, reward modeling, and computational cognitive science.

Expected Sep 2026

### Indiana University

*M.S. in Computer Science*

2018–2020

### Indiana University

*B.S. in Informatics, B.A. in Mathematics*

2011–2016

## Industry Research

### Radical Numerics

*Member of Technical Staff Intern*

Oct 2025–Jan 2026

- Built an end-to-end post-training and evaluation stack for DPO-tuning genomic foundation models, including preference-data processing, training orchestration, and evaluation harnesses for biomedical DNA-sequence design.

### Toyota Research Institute

*Research Scientist Intern, Human-AI Interactive Learning*

Jun-Sep 2025

- Built a teacher-training objective for corrective feedback in which teacher outputs were scored by how much they raised a frozen student's likelihood of the reference solution.
- Established a reliable final-answer reward signal and characterized why it failed to transfer to long reasoning traces.

## Publications

\*Equal contribution.

### LLM Post-Training, Reward Modeling, and Evaluation

**V Xiang**, M Khalifa, I Wu, Y Shen, J Leskovec, N Haber, A Kumar. Test-Time Self Improvement by Scaling Unsupervised World Internalization. *Under review at ICLR 2027*.

- Developed U-WIN: an agent explores a new environment, writes notes on what it learns, and trains on those notes plus self-generated trajectories that apply them, with no access to downstream tasks, rewards, or correctness feedback.
- Showed downstream performance improves near log-linearly from 10M to 100M training tokens across legal, codebase, and math environments (Harvey, StudyBench, Equational Theories) with Qwen 9B and 27B models.
- Raised the share of agentic trajectories that recall environment knowledge from 20.6% to 99.8%; self-written notes also beat token-matched raw documents or observations for knowledge retention.

**V Xiang**, A Setlur, C Blagden, N Haber, A Kumar. [ExpRL: Exploratory RL for LLM Mid-Training](#). *COLM 2026* | [code](#).

- Designed and open-sourced ExpRL: an RL-based mid-training method using reference solutions as reward scaffolds for an LLM judge that scores partial progress with dense outcome- and process-level rewards for online RL.
- Beat the strongest baseline at each stage on held-out AIME 2026 pass@1 (Qwen3-4B) by 3.6 points after ExpRL priming alone and by 4.7 points after identical sparse-GRPO post-training.

**D Fein\***, M Lamparth\*, **V Xiang**, M J Kochenderfer, N Haber. [One Bias After Another: Mechanistic Reward Shaping and Persistent Biases in Language Reward Models](#). *ICML 2026* | [code](#).

- Identified five reward-model biases (length, uncertainty, position, sycophancy, model-style) across leading Bradley-Terry (BT) RMs.
- All five biases persist: 22.6-point accuracy drops on hedged correct answers, up to 60% agreement with wrong user suggestions, and positional swings up to 28 points.
- Reduced the length, uncertainty, and position biases via null-space probe projections without retraining or RewardBench-2 loss.

**D Fein\***, S Russo\*, **V Xiang\***, K Jolly, R Rafailov, N Haber. [LitBench: A Benchmark and Dataset for Reliable Evaluation of Creative Writing](#). *EACL 2026, Long Paper* | [dataset](#).

- Curated LitBench, the first standardized benchmark for creative-writing evaluation: a 43k-pair training corpus and a 2.5k-pair debiased, human-labeled test set.
- Trained BT and generative reward models that beat the strongest zero-shot judge by 5 points in human-preference agreement.
- Validated reward-model rankings via an online human study on newly LLM-generated stories.

**V Xiang**, C Blagden, R Rafailov, N Lile, S Truong, C Finn, N Haber. [Just Enough Thinking: Efficient Reasoning with Adaptive Length Penalties Reinforcement Learning](#). *NeurIPS 2025 Efficient Reasoning Workshop Spotlight*.

- Developed a GRPO-compatible adaptive length penalty for DeepScaleR-1.5B that reduced generation length by more than 50% at matched accuracy across MATH-500, AIME, and OlympiadBench while allocating  $5.35\times$  more tokens to hard than easy prompts.

**T Hua**, H Hua, **V Xiang**, B Klieger, S Truong, W Liang, F-Y Sun, N Haber. [ResearchCodeBench: Benchmarking LLMs on Implementing Novel Machine Learning Research Code](#). *NeurIPS 2025 Datasets & Benchmarks Track, Spotlight* | [website](#).

- Co-developed an open-source execution harness for 212 challenges from 20 recent ML papers; across 32 evaluated models, the strongest reached 33.0% on the HARD split, and 43 tasks remained unsolved by every model.

**V Xiang**, C Snell, K Gandhi, A Albalak, A Singh, C Blagden, D Phung, R Rafailov, N Lile, D Mahan, L Castricato, J-P Franken, N Haber, C Finn. [Towards System 2 Reasoning in LLMs: Learning How to Think With Meta Chain-of-Thought](#). *Preprint*, 2025.

- Proposed a training roadmap combining linearized search traces, process supervision, synthetic data, and reinforcement learning, supported by evidence of in-context search behavior in frontier reasoning models.

## Multi-Agent Learning

L Cross, **V Xiang**, A Bhatia, D L K Yamins, N Haber. [Hypothetical Minds: Scaffolding Theory of Mind for Multi-Agent Tasks with Large Language Models](#). *ICLR 2025, Poster*.

- Contributed to the design and implementation of an LLM Theory-of-Mind agent that outperformed MARL baselines on 3 of 4 environments and 6 of 8 eight-agent scenarios across 30 Melting Pot tasks.

**V Xiang**, L Cross, J-P Fränken, N Haber. [From Centralized to Self-Supervised: Pursuing Realistic Multi-Agent Reinforcement Learning](#). *NeurIPS 2023 ALOE Workshop, Poster*.

- Built and compared centralized, decentralized, asynchronous, and self-supervised MARL pipelines, showing how centralized-information assumptions can overstate performance in realistic mixed-motive settings.

## Human and Machine Learning

B Long\*, R Sparks\*, **V Xiang\***, S Stojanov\*, et al.. [The BabyView Dataset: High-Resolution Egocentric Videos of Infants' and Young Children's Everyday Experiences](#). *Computational Cognitive Neuroscience 2025 Proceedings*.

- Co-developed an 868-hour longitudinal egocentric-video dataset spanning ages 5 months to 3 years, with gold-standard annotations and evaluation pipelines for speech transcription, diarization, pose estimation, and out-of-distribution transfer.

C Zhuang\*, **V Xiang\***, Y Bai, X Jia, N Turk-Browne, K Norman, J J DiCarlo, D L K Yamins. [How Well Do Unsupervised Learning Algorithms Model Human Real-Time and Lifelong Learning?](#). *NeurIPS 2022 Datasets & Benchmarks Track*.

- Developed naturalistic real-time and lifelong visual-learning benchmarks and identified memory mechanisms as critical for learning from sparse, low-diversity input streams.

B Long, S Goodin, G Kachergis, V A Marchman, S F Radwan, R Z Sparks, **V Xiang**, et al.. [The BabyView Camera: Designing a New Head-Mounted Camera to Capture Children's Early Social and Visual Environments](#). *Behavior Research Methods* 56, 2024.

L Yuan, **V Xiang**, D Crandall, L Smith. [Learning the Generative Principles of a Symbol System from Limited Examples](#). *Cognition*, 2020.

## Privacy and Security

Y Liu, **V Xiang**, E J Seong, A Kapadia, D S Williamson. [Defending Against Microphone-Based Attacks with Personalized Noise](#). *Proceedings on Privacy Enhancing Technologies*, 2021.

## Technical Skills

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**Library & Tools:** FSDP, Docker, vLLM, SGLang, Ray/RLlib, SLURM

**Deep Learning Frameworks:** PyTorch, Tensorflow, Keras

**Programming Languages:** Python, C++, C, JavaScript

## Teaching

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|---------|----------------------------------------------------------------------------------------|
| 2023-25 | Teaching Assistant, Neural Network Models of Cognition, Stanford University            |
| 2023    | Teaching Assistant, Introduction to Statistical Methods, Stanford University           |
| 2021    | Teaching Assistant, Longitudinal Design and Data Analysis, Stanford University         |
| 2019    | Mentor, Proactive Health Informatics REU, Indiana University                           |
| 2019    | Teaching Assistant, Computer Vision, Indiana University                                |
| 2018    | Teaching Assistant, Introduction to Data Structures and Algorithms, Indiana University |